

SYSTEM DESIGN DOCUMENT: INERTIAL THRUSTER ECONOMY

Project Title: Astro-Logic Movement System

Document Version: 1.0

Status: Implementation Ready



1. DESIGN PILLARS & CORE CONCEPT

The Inertial Thruster System is a resource-driven movement engine designed for a 2D platformer. Unlike standard platformers where movement is a constant, this system introduces a **Resource-Agility Trade-off**.

- **Tactical Scarcity:** Fuel is a limited currency used for verticality and hovering.
- **Variable Physics:** The player's mass fluctuates based on fuel levels, altering the "feel" of movement.
- **Risk/Reward Loops:** Low fuel grants higher agility but removes the "safety net" of thrusters.

2. THE RESOURCE LOOP (FUEL)

The core of the system is the **Fuel Tank (100 Units)**. It acts as both a power source and a physical weight.

A. Consumption (Sinks)

- **High Jump:** Consumes 10% fuel instantly. Increases jump height from 2 units to 5 units.
- **Hovering:** Consumes 5% per second. Allows the player to maintain Y-axis altitude.
- **The Sputter (Fail-safe):** At 0% fuel mid-air, a 2-unit horizontal impulse is applied as a final "lifeboat" before thrusters are disabled.

B. Replenishment (Sources)

- **Auto-Refill:** 2% per second. Requires the player to be Grounded for 2 seconds without jumping. Any jump reset the timer.
- **Gas Stands:** Instant 100% refill. Operates on a 10-second per-object cooldown to prevent stationary hovering abuse.

3. VARIABLE PHYSICS REGISTRY (The Math)

The "feel" of the game is driven by an inverse relationship between Fuel and Agility.

Fuel Load %	Player Weight	Jump Height	Jump distance	Move Speed
0%	$W - 0.10 \cdot W$	$Y + 0.25 \cdot Y$ m (2.5 units)	$X + 0.25 \cdot X$ (5 units)	$V + 0.25 \cdot V$ (5.25 units)
10% 	W (91 kg)	2.45 units	4.9 units	5.225 m/s
20%	W (92 kg)	2.40 units	4.8 units	5.20 m/s
30%	W (93 kg)	2.35 units	4.7 units	5.175 m/s
40%	W (94 kg)	2.30 units	4.6 units	5.15 m/s
50%	W (95 kg)	2.25 units	4.5 units	5.125 m/s
60%	W (96 kg)	2.20 units	4.4 units	5.10 m/s
70%	W (97 kg)	2.15 units	4.3 units	5.075 m/s
80%	W (98 kg)	2.10 units	4.2 units	5.05 m/s
90%	W (99 kg)	2.05 units	4.1 units	5.0125 m/s
100%	W (100 kg)	Y (2 units)	X (4 units)	V (5 m/s)

Design Intent: Skills like the "Light Jump" (2.25 units) allow expert players to clear platforms that are impossible at 100% fuel, encouraging intentional fuel burning.

4. STATE MACHINE LOGIC

The system handles transitions between four primary states to ensure "Juice" and stability:

- **Grounded State:** Checks for the 2s delay. Triggers "Engine Prime" VFX/SFX.
- **Powered Flight:** Checks Fuel > 10%. Disables auto-refill logic.
- **Sputter State:** Triggered at Fuel = 0 in-air. Disables thruster input until IsGrounded = True.
- **Refilling State:** Active only when IsGrounded and Timer > 2s.

5. LEVEL DESIGN CONSTRAINTS (Atoms)

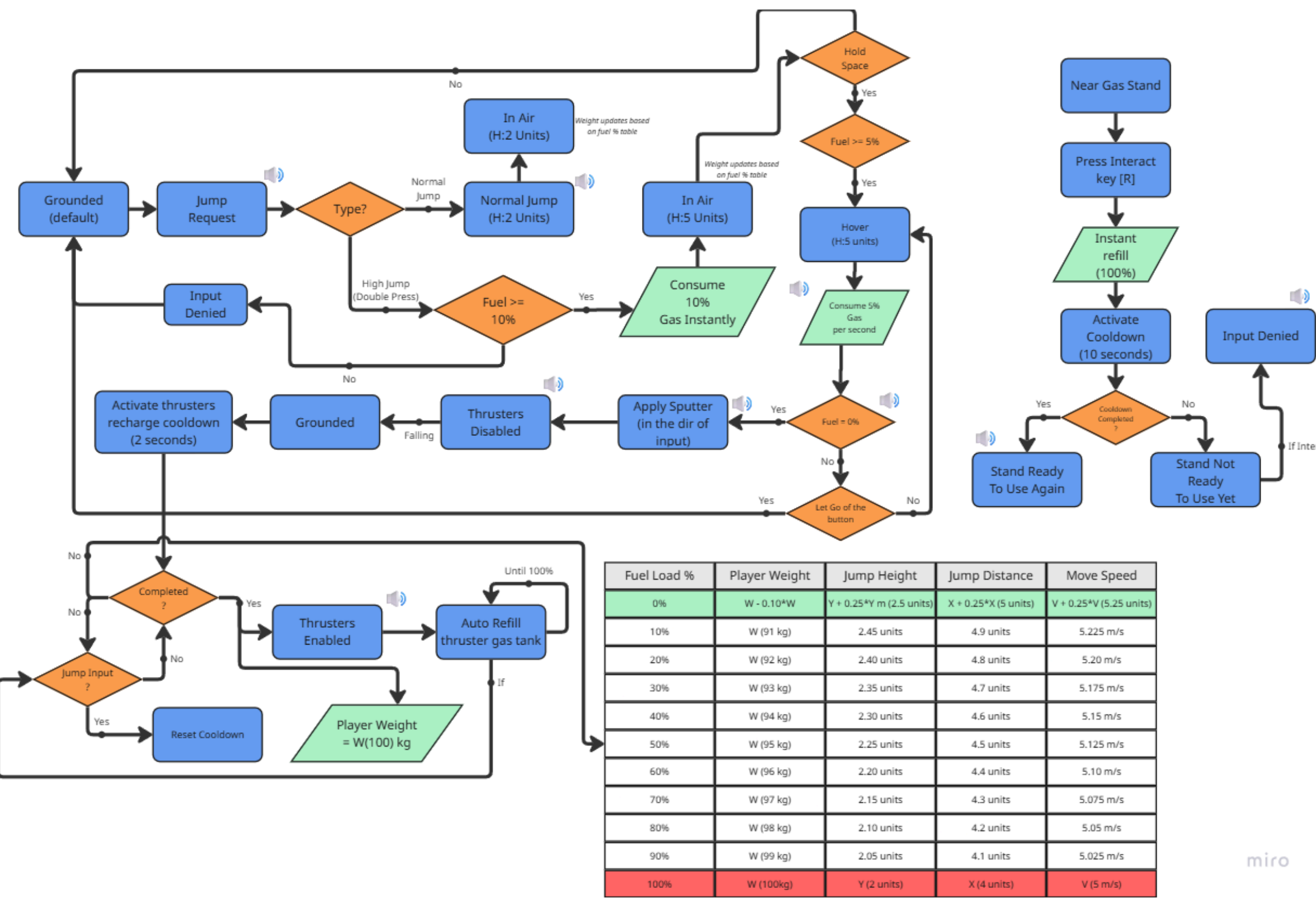
Based on the system math, the following level design "Rules of Truth" are established:

- **The Hard Gate:** Any wall > 5 units requires a helper platform (Jump cap is 5).
- **The Agility Gap:** A 2.2-unit ledge is a "Weight Gate"—it can only be reached if the player has < 15% fuel.
- **The Recovery Zone:** Platforms must be at least 4 units wide to safely allow the 2s Grounded Refill without falling off.

6. UX & FEEDBACK (The Juice)

- **Visuals:** Thrusters shake and emit "dark smoke" during the 2s grounding phase.
- **Audio:** Low Fuel: High-pitched alarm at < 10%.
- **Sputter:** Mechanical "coughing" sound.
- **Refill Success:** Short sci-fi "ping" after the 2s delay ends.

SYSTEM DESIGN FLOW



**Speaker icons represent that there must be an appropriate sound effect when the respective state comes.*